1. Difference Between Frontend, Backend, and Full-Stack Development (with Examples)

- **Frontend Development**: Focuses on the *client-side* — what users see and interact with in a browser.
  - *Languages*: HTML, CSS, JavaScript.
  - *Example*: A shopping website's layout, buttons, product images.
- **Backend Development**: Focuses on the *server-side* — where logic, database operations, and authentication occur.
  - *Languages*: Python, Node.js, PHP, Java.
  - *Example*: User login validation, storing orders in a database.
- **Full-Stack Development**: Combines both frontend and backend.
  - *Example*: A full-stack dev can build a blog from scratch — designing the UI and also building APIs to save posts.

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2. Simple Diagram of Client-Server Model

User (Client)

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[Browser]

↓ HTTP Request

[Web Server]

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[Application Logic]

↓

[Database Server]

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↑

[Web Server]

↑ HTTP Response

[Browser]

↑

User (Client)

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3. How a Browser Requests and Displays a Web Page

1. User enters a URL (e.g., www.example.com).
2. DNS converts the URL into an IP address.
3. Browser sends an HTTP request to the web server.
4. Server processes it and sends back an HTTP response with HTML/CSS/JS.
5. Browser renders the content using rendering engine (e.g., Blink in Chrome).

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4. Tools Required for Web Development & Their Purpose

Tool	Purpose
VS Code	Code editor with extensions for web tech
Browser Dev Tools	Debug frontend code (HTML/CSS/JS)
Node.js	Run JavaScript backend, manage packages
Git & GitHub	Version control and collaboration
Live Server Extension	Auto-reloads browser during dev

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5. What is a Web Server? With Examples

- A **web server** handles requests from the client (browser) and serves web pages or data.
- It can host HTML files, handle API requests, or run server-side logic.

Examples:

- Apache
- Nginx
- Microsoft IIS
- Node.js (Express-based servers)

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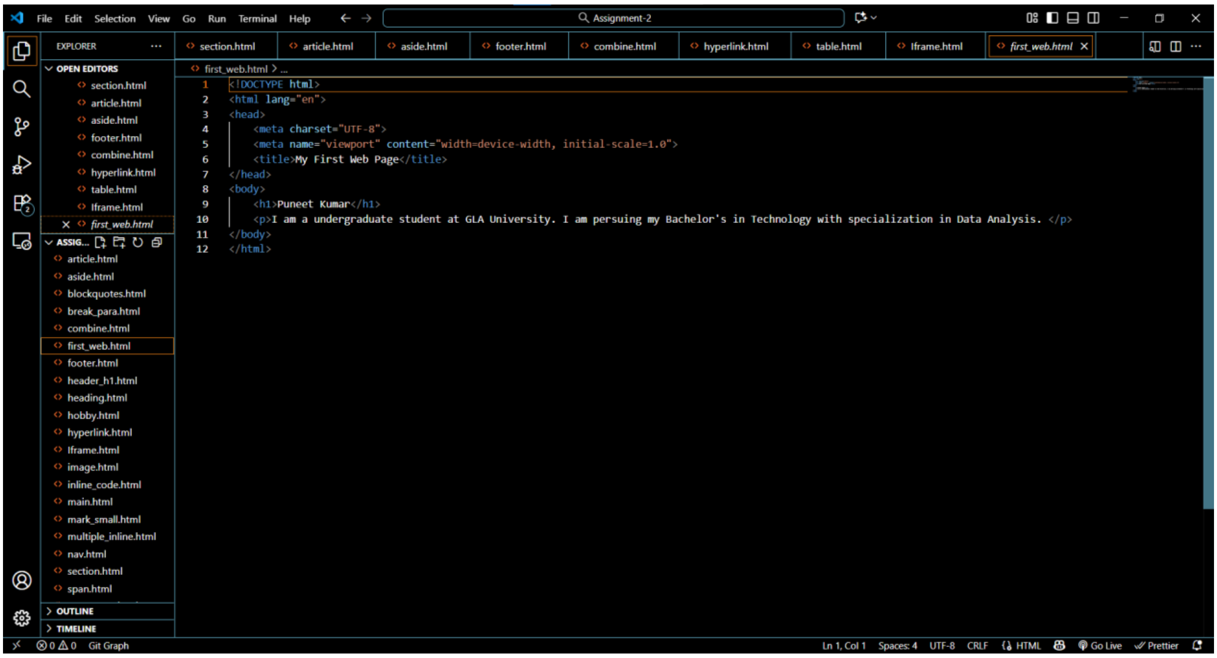
6. Roles in a Web Development Project

Role	Responsibility
Frontend Developer	Designs and builds UI with HTML/CSS/JavaScript
Backend Developer	Builds server logic, handles database operations and APIs
Database Administrator (DBA)	Manages database structure, performance, security

7. **Configure VS Code for web development**



Let's dive in

Life is beautiful

Life's getting updated (smile more)

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8. Static vs Dynamic Websites

Type	Static Website	Dynamic Website
Content	Fixed content	Changes based on user or data
Tech	HTML/CSS	HTML + Backend (PHP, Node.js, etc.)
Example	Portfolio page	Facebook, YouTube

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9. Five Web Browsers & Rendering Engines

Browser	Rendering Engine
Google Chrome	Blink
Mozilla Firefox	Gecko
Safari	WebKit
Microsoft Edge	Blink (after switching from EdgeHTML)
Opera	Blink

Rendering Engine is responsible for parsing HTML/CSS and displaying it. They differ in:

- Speed
- Support for new features
- Developer tools
- Performance optimizations

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10. Diagram: Web Architecture (Client → Server → DB)

[Client (Browser)]

↓ Request

[Web Server (Apache/Nginx)]

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[Application Server (Node.js/Python)]

↓

[Database (MySQL/MongoDB)]

↑

↕ (API calls)

APIs (REST or GraphQL) allow clients to talk to servers and access data from databases.